

FUN19 (P28003 / YAL034C) —

Transcription Coactivation After Ada2/ SAGA Divergence

Focus type: function_assignment **Hypothesis slug:** transcription-coactivation-after-ada2-saga-divergence **Target:** *Saccharomyces cerevisiae* FUN19 (UniProt P28003; systematic name YAL034C)

Executive Judgment

Verdict: REFUTED / OVER-ANNOTATED.

The seed hypothesis proposes that FUN19 has **transcription coactivator activity** (GO:0003713) and **positively regulates DNA-templated transcription** (GO:0045893), reasoning that although FUN19 lost the Ada2 ZZ/SANT domains and SAGA-complex membership, it retained SWIRM and therefore may keep "some activation mechanism." The evidence assembled across three investigative iterations does not support this. Every strand of evidence — domain architecture, existing GO annotation set, and both aggregated (STRING) and curated (IntAct) interaction data — points away from positive coactivation and, where it points anywhere, points toward a **repressive (Rpd3L/Sin3 histone-deacetylase) context** or toward **chaperone/nuclear-import associations** that have nothing to do with transcriptional activation.

The core logical flaw in the seed hypothesis is that it treats the retained **SWIRM domain** as a residual activation module. Primary structural work on the SAGA HAT module shows the opposite: SWIRM is a **tethering/interaction module**, not a catalytic or intrinsically coactivating element. The domains that actually confer Ada2's coactivator function — the ZZ-type zinc finger and the SANT domain, which stabilize Gcn5 and the histone-acetyltransferase reaction — are exactly the domains FUN19 has **lost**. Retaining the "docking" piece while losing the "business end" is not a conservative basis for inferring retained activation.

The most important caveats: (1) FUN19 is an essentially **uncharacterized, non-essential gene** with no experimental FUNCTION or SUBUNIT annotation in UniProt, so the case is built on comparative/interaction inference rather than a direct assay that refutes coactivation; (2) all four current GO annotations are **IBA (phylogenetic inference, GO_Central)** with no experimental backing, so the annotation baseline itself is weak in both directions; and (3) the STRING SIN3 signal, while the strongest experimental partner, is aggregated high-throughput co-occurrence rather than a validated binary complex membership. The recommendation is therefore to **reject the coactivator / positive-regulation assignment** and retain only the neutral IBA-level chromatin terms pending direct experimental evidence.

Key Findings

F001 — FUN19 retained only SWIRM and lost the Ada2 domains that confer coactivator function

FUN19 (UniProt P28003; YAL034C; 413 aa) carries a **single annotated structured domain — SWIRM (residues 316–413; Pfam PF04433; PROSITE PS50934)** — plus three disordered/low-complexity regions. It has **no HAT domain, no bromodomain, no ZZ-type zinc finger, and no SANT domain**. This is in direct architectural contrast to yeast ADA2 (UniProt Q02336; YDR448W; 434 aa), which carries a **ZZ-type zinc finger (residues 2–58), a SANT domain (60–112), and a SWIRM domain (349–434)**. ADA2 is one of the ~19 subunits of the SAGA coactivator complex (TRA1, SPT7, TAF5, NGG1/ADA3, ADA2, GCN5, ...). **FUN19 is not in the SAGA subunit list**. Both proteins fall within PANTHER family PTHR12374 (ADA2-related), but FUN19 sits in the distinct **subfamily SF21 (FUN19-related)**, consistent with an ancient duplication/divergence event rather than orthology to functional Ada2.

The mechanistic significance of this domain loss is established by primary structural work on the SAGA HAT module ([PMID: 36965704](#)):

"A minimal HAT module complex containing Gcn5 bound to Ada2 and Ada3 is required for full Gcn5 activity on nucleosomes."

This establishes that Ada2's coactivator (histone-acetyltransferase) role depends on the assembled Ada2/Ada3/Gcn5 module. The domains Ada2 uses to build and stabilize that module — beyond SWIRM — are precisely the ZZ and SANT domains FUN19 lacks. The same paper further clarifies what SWIRM actually does:

"the SWIRM domain is required for incorporation of the HAT module into the yeast SAGA complex, but not the ADA complex."

SWIRM is therefore a **tethering/interaction module** that helps dock the HAT module into SAGA — it is not a catalytic or intrinsically activating element. Its presence in FUN19 does **not** by itself imply any coactivator activity. The seed hypothesis's premise ("retains SWIRM; those losses may change complex context without eliminating every activation mechanism") is thus undermined at the structural level: FUN19 kept the docking peg and discarded the machine.

F002 — Existing FUN19 GO annotations are neutral or repressive in context and are entirely IBA

All four current FUN19 (P28003) GO annotations are **IBA:GO_Central** — phylogenetic inference from GO Central, with **no experimental support**:

Aspect	GO ID	Term	Directionality
CC	GO:0070210	Rpd3L-Expanded complex	Repressive (HDAC)
MF	GO:0003682	chromatin binding	Neutral
BP	GO:0006338	chromatin remodeling	Neutral
BP	GO:0006357	regulation of transcription by RNA Pol II	Direction-neutral

Critically, **neither GO:0003713 (transcription coactivator activity, MF) nor GO:0045893 (positive regulation of DNA-templated transcription, BP) is currently annotated to FUN19**. The seed hypothesis is proposing to *add* activation-directional terms that the existing curated set does not contain. Moreover, the one complex context that *was* propagated — **Rpd3L-Expanded (GO:0070210)** — is a **histone-deacetylase (repressive) assembly**, i.e., opposite in sign to "positive" coactivation. UniProt lists **no experimental FUNCTION or SUBUNIT comment** for FUN19 (only that it is a non-essential gene and a phosphoprotein). The existing annotation baseline therefore offers no support for adding activation directionality and, if anything, its only complex-level context leans repressive.

F003 — FUN19's strongest experimental interaction is SIN3 (co-repressor scaffold); SAGA links are database carry-over

STRING (*S. cerevisiae*, YAL034C) interaction partners were examined by evidence channel. The highest **experimental-evidence (escore)** partners are the **co-repressor scaffold SIN3 (escore = 0.755)**, KAP123 (0.727), TAF1 (0.668), ADH1 (0.625), and TAF5 (0.608). By contrast, **all recovered SAGA subunits** — GCN5, ADA2, NGG1/ADA3, SPT7, SPT20, SPT3, SPT8, TRA1, SGF29, SGF73, SUS1, HFI1 — have **escore = 0.000** and a uniform **dscore = 0.540**, meaning they derive from **database/complex co-annotation, not from any experiment**. In other words, the apparent "SAGA membership" that seeds the coactivator hypothesis is a **database artifact**: it is co-annotation transfer, not experimental co-purification. Among curated complex subunits, FUN19's only genuine experimental interaction signal maps to **SIN3**, the scaffold of the **Rpd3L/Rpd3S co-repressor** — again a repressive, not activating, context.

F004 — Curated IntAct interactions place FUN19 with chaperones and nuclear-import factors, not any coactivator complex

Curated IntAct/PSICQUIC experimental interactions for P28003 are **dominated by molecular chaperones and folding machinery**: Ssa1 (Hsp70), Hsc82 (Hsp90), Hsp104, Sse2, Gim4 (prefoldin), Aha1, and Csa1 — recovered via tandem affinity purification in the chaperone interaction network ([PMID: 19536198](#)) — plus **KAP123** (karyopherin/nuclear import; TAP, [PMID: 16429126](#)) and a two-hybrid self-association ([PMID: 14690591](#)). There is **no curated FUN19–Sin3 or FUN19–SAGA binary interaction** in IntAct. This has two implications. First, it further confirms that FUN19 is **not a validated member of any transcription coactivator complex**. Second, it tempers the STRING SIN3 signal: the escore of 0.755 reflects **aggregated high-throughput co-occurrence** rather than an established stable complex, so even the repressive-context lead should be treated as **suggestive, not proof** of Rpd3L subunit status. The chaperone-heavy interactome is more consistent with a protein that is metastable/poorly folded on its own, or that is captured non-specifically by folding machinery, than with a stably assembled transcriptional coactivator.

Mechanistic Model / Interpretation

The seed hypothesis and the evidence can be summarized in a single comparative picture:

	ZZ finger	SANT	SWIRM	HAT module role
ADA2 (Q02336) (SAGA subunit)	YES	YES	YES	Coactivator: stabilizes Gcn5, enables full HAT activity on nucleosomes; SWIRM docks HAT module into SAGA
FUN19 (P28003) (subfamily SF21)	NO	NO	YES	Docking peg only; no catalytic or activation domains retained

The logical chain the seed hypothesis relies on is: *SWIRM retained* → *some activation mechanism survives Ada2/SAGA divergence* → *assign coactivator + positive regulation*. Each link fails:

1. **SWIRM is not an activation module.** Structural work shows SWIRM tethers the HAT module into SAGA; it does not carry out or intrinsically drive acetylation/activation (P 36965704).

Retaining it does not retain activation.

2. **The activating domains were lost.** The ZZ and SANT domains — those that stabilize Gcn5 and enable full HAT activity — are absent from FUN19. Loss of the effector domains is the functionally decisive change, not the retained docking module.
3. **The complex context that survived is repressive.** The only propagated complex term is Rpd3L-Expanded (an HDAC assembly), and the only genuine experimental interaction among transcription factors is SIN3 (the Rpd3 co-repressor scaffold). If FUN19 has any residual chromatin role, the vector of evidence points **repressive**, not activating.
4. **The "SAGA membership" is an artifact.** STRING SAGA links have experimental score 0.000; they are co-annotation transfer. IntAct records no SAGA/Sin3 binary interaction at all.

The coherent interpretation is that FUN19 is a **divergent, largely uncharacterized paralog** within the ADA2 (PTHR12374) family that **lost the coactivator effector apparatus**. Its curated interactome (chaperones + a karyopherin) is that of a protein whose stable partners are folding/transport machinery, not a transcriptional activation module. Any transcription-related role is at best **neutral chromatin association**, and the strongest specific lead is a **repressive Sin3/Rpd3 context** — the opposite sign from the hypothesis.

GO Decision Table (lead requiring curator verification)

GO term	Aspect	Seed proposes	Evidence status	Recommended action
GO:0003713 transcription coactivator activity	MF	ADD	No experimental support; effector domains lost; interactome repressive/ chaperone	Do not add (reject)
GO:0045893 positive regulation of DNA-templated transcription	BP	ADD	No experimental support; only complex context is repressive HDAC	Do not add (reject)
GO:0070461 SAGA-type complex (loss/IRD)	CC	(context)	SAGA links are DB artifact (escore 0.000); FUN19 not a SAGA subunit	Consistent with no SAGA membership
GO:0070210 Rpd3L-Expanded complex	CC	(retained)	IBA only; SIN3 is top experimental partner but HT co-occurrence	Retain as IBA/non-core , flag as unverified
GO:0003682 chromatin binding	MF	(retained)	IBA only	Retain as IBA, neutral
GO:0006338 chromatin remodeling	BP	(retained)	IBA only; does not require ATPase activity of target	Retain as IBA, neutral
GO:0006357 regulation of transcription by Pol II	BP	(retained)	IBA only; direction-neutral	Retain as IBA, neutral

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
UniProt P28003 (FUN19) vs Q02336 (ADA2)	Structural / domain architecture	Refutes	FUN19 retains Ada2's coactivator machinery	FUN19 has only SWIRM (316–413); lacks ZZ, SANT, HAT, bromodomain that ADA2 carries	<i>S. cerevisiae</i> protein annotation	High for architecture; inference for function
PMID: 36965704	Direct structural/ biochemical	Refutes (mechanism)	SWIRM is an activation module	SWIRM is a tethering module (docks HAT into SAGA); coactivation needs Ada2/ Ada3/Gcn5 module	Yeast SAGA HAT module	High; about ADA2, applied to FUN19 by analogy
Current GO annotation set (GO_Central IBA)	Review/ database	Refutes / qualifies	Coactivator/ positive terms already supported	No GO:0003713 or GO:0045893; only IBA terms; one context is repressive Rpd3L	<i>S. cerevisiae</i>	High for annotation state; IBA has no experimental weight
STRING YAL034C	Interaction (aggregated)	Refutes / qualifies	FUN19 is a SAGA coactivator member	SAGA subunits escore=0.000 (DB carry-over); top experimental partner is SIN3 (escore	<i>S. cerevisiae</i>	Medium; SIN3 signal is HT co-occurrence, not binary

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context	Confidence & limitations
				0.755, co-repressor)		
PMID: 19536198	Interaction (curated, TAP)	Refutes / competing	FUN19 sits in a transcription coactivator complex	FUN19 interactome dominated by chaperones (Ssa1, Hsc82, Hsp104, prefoldin)	<i>S. cerevisiae</i> chaperone network	High for interaction; suggests folding-machinery association
PMID: 16429126	Interaction (curated, TAP)	Qualifies	FUN19 nuclear-transcriptional role	FUN19–KAP123 (karyopherin/import) interaction	<i>S. cerevisiae</i>	Medium; consistent with nuclear localization, not activation
PMID: 14690591	Interaction (curated, Y2H)	Qualifies	—	FUN19 self-association	<i>S. cerevisiae</i> two-hybrid	Low functional weight

Evidence Base (Literature)

PMID: 36965704 — *The SAGA HAT module is tethered by its SWIRM domain and modulates activity of the SAGA DUB module.* This is the pivotal mechanistic reference. It directly establishes (1) that full Gcn5/HAT activity on nucleosomes requires the assembled Gcn5–Ada2–Ada3 module, and (2) that the SWIRM domain is a **tethering** element that incorporates the HAT module into SAGA rather than a catalytic/activating element. Both quotes were verified against the abstract. Together they dismantle the seed hypothesis's central premise that retention of SWIRM implies retention of coactivation: the domain FUN19 kept is the docking module, while the domains it lost (ZZ, SANT) are those coupled to Gcn5 stabilization and HAT activity.

PMID: 19536198 (chaperone interaction network, TAP-MS) — source of the curated FUN19 interactions with Hsp70/Hsp90/Hsp104/prefoldin. Supports the interpretation that FUN19's stable curated partners are folding machinery, not a coactivator complex.

PMID: 16429126 (TAP) — FUN19–KAP123 (nuclear import) interaction. Consistent with nuclear localization but not with transcriptional activation.

PMID: 14690591 (Y2H) — FUN19 self-association; low functional weight but part of the curated interaction record.

Database-level orientation (STRING, IntAct/PSICQUIC, UniProt, PANTHER PTHR12374) is used as orientation, not as primary support, per the research directive — the STRING escore breakdown is treated as evidence of **artifactual** SAGA co-annotation rather than as positive interaction evidence.

Conflicts and Alternatives

1. **Paralog over-annotation (most likely explanation).** FUN19 sits in PANTHER PTHR12374 (ADA2 family) but in a distinct subfamily (SF21). IBA propagation from the ADA2-containing family, plus STRING/complex co-annotation, generates the appearance of SAGA/coactivator association. The escore=0.000 for every SAGA subunit is the signature of database carry-over, not biology.
2. **Repressive alternative.** If FUN19 has any real chromatin function, the evidence sign is **repressive**: the only propagated complex is Rpd3L-Expanded (HDAC), and the top experimental interactor is the Sin3 co-repressor scaffold. This is the opposite of the seed hypothesis and is the more parsimonious residual-function lead.
3. **Non-transcriptional / chaperone-client alternative.** The curated IntAct interactome (Hsp70/Hsp90/Hsp104/prefoldin/karyopherin) is equally consistent with FUN19 being a poorly characterized, possibly metastable protein whose principal validated partners are folding and transport machinery — not a transcription factor at all.
4. **Weak baseline in both directions.** Because every GO annotation is IBA and there is no experimental FUNCTION comment, no interpretation is anchored on direct assay. The refutation is strong relative to the *added* claim (coactivation) but should not be overstated into a positive assignment of repressive function.

Limitations and Knowledge Gaps

- **No direct functional assay for FUN19.** There is no reporter, ChIP, or transcriptome study establishing FUN19's effect on transcription in either direction. The case against coactivation is comparative and interaction-based, not a direct refuting experiment. *Resolution:* an inducible-reporter or nascent-transcription (e.g., 4tU-seq) assay in a *fun19Δ* strain.
 - **STRING SIN3 signal is aggregated, not binary.** The escore of 0.755 is high-throughput co-occurrence; IntAct records no FUN19–Sin3 binary interaction. Whether FUN19 is a genuine Rpd3L/Sin3 subunit is unresolved. *Resolution:* co-IP / reciprocal TAP of Sin3 vs FUN19; size-exclusion co-migration.
 - **SWIRM functional test in FUN19 context.** SWIRM tethers the HAT module in ADA2; whether FUN19's SWIRM binds any chromatin-modifying module is untested. *Resolution:* pulldown of FUN19-SWIRM against HAT/HDAC modules.
 - **Localization confirmation.** KAP123 interaction suggests nuclear import but a direct GFP localization for FUN19 was not established in this analysis. *Resolution:* FUN19-GFP microscopy (or consult existing SGD localization data).
 - **PANTHER IRD / ancestral judgment.** The seed cites an IRD loss of SAGA-type-complex membership (GO:0070461) at the fungal node. This was interpreted from the seed context, not independently re-derived from the PANTHER tree in these iterations. *Resolution:* directly inspect the PTHR12374 tree/IRD annotations.
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Discriminating Tests (most efficient)

1. ***fun19Δ* nascent-transcription / reporter assay.** A directional readout: if FUN19 were a coactivator, its deletion should decrease activated transcription. This is the single most decisive test to distinguish coactivation from a repressive or null role.
2. **Reciprocal co-IP of Sin3 and Gcn5/Ada2 with FUN19.** Distinguishes genuine Rpd3L/Sin3 (repressive) membership from artifactual SAGA (activating) co-annotation. Predicts Sin3 pulls down FUN19 while SAGA does not.
3. **FUN19-SWIRM biochemical pulldown vs HAT and HDAC modules.** Tests whether the retained SWIRM tethers FUN19 to any chromatin-modifying complex, and if so which sign.

4. **FUN19-GFP localization + KAP123-dependence.** Confirms nuclear vs cytoplasmic partitioning and whether the karyopherin interaction is functional.
5. **Comparative PANTHER/IRD re-derivation.** Independently confirm the ancestral judgment (SAGA-membership loss at the fungal node) rather than relying on the seed's summary.

Curation Leads (require curator verification)

Primary lead — reject the added activation terms. - **Do NOT add GO:0003713 (transcription coactivator activity, MF).** No experimental support; Ada2 effector domains (ZZ, SANT) are lost; SWIRM is a tether, not an activator; interactome is repressive/chaperone-dominated. - **Do NOT add GO:0045893 (positive regulation of DNA-templated transcription, BP).** Same reasoning; the only propagated complex context is the repressive Rpd3L/Sin3 HDAC assembly.

Secondary leads — annotation hygiene. - **Retain the existing IBA terms as non-core / unverified:** GO:0070210 (Rpd3L-Expanded), GO:0003682 (chromatin binding), GO:0006338 (chromatin remodeling), GO:0006357 (regulation of transcription by Pol II). Flag each as IBA-only, no experimental support. - **Flag SAGA-membership propagation as artifactual.** Any pipeline that would transfer SAGA/coactivator terms to FUN19 from PTHR12374 should be blocked; STRING SAGA escore = 0.000 confirms these are co-annotation carry-over; and PANTHER records an IRD loss of SAGA-type-complex membership at the fungal node. - **Note the repressive lead, but do not annotate it as fact.** The SIN3 (escore 0.755) association is the strongest experimental transcription-factor link and, with GO:0070210, suggests a repressive context — but IntAct shows no binary Sin3–FUN19 interaction, so this is a lead, not an assignment.

Candidate references with snippets to verify: - [PMID: 36965704](#): *"the SWIRM domain is required for incorporation of the HAT module into the yeast SAGA complex, but not the ADA complex"* — supports SWIRM = tether, not activator. - [PMID: 36965704](#): *"A minimal HAT module complex containing Gcn5 bound to Ada2 and Ada3 is required for full Gcn5 activity on nucleosomes."* — supports that coactivation requires the effector module FUN19 lacks.

Suggested curator questions: - Is there any experimental (non-IBA) evidence anywhere in the literature for FUN19 affecting transcription in either direction? - Should FUN19's IBA Rpd3L-Expanded annotation be retained given no binary Sin3 interaction in IntAct?

Conclusion

The seed hypothesis that FUN19 has transcription coactivator activity and positively regulates transcription is **refuted** / **best treated as paralog-driven over-annotation**. FUN19 retained only Ada2's SWIRM tethering module while losing the ZZ and SANT domains that underpin coactivation; it is not a SAGA subunit; all current annotations are IBA; the only propagated complex context and the strongest experimental interactor are **repressive** (Rpd3L / Sin3), and curated interactions place FUN19 with chaperones and a karyopherin rather than any coactivator complex. Curators should **not** add GO:0003713 or GO:0045893 and should retain only the neutral IBA-level chromatin terms pending direct experimental validation.

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